Sandeep Vaid

Backend engineer with 4+ years of building scalable data pipelines, cloud-native APIs, and full-stack products. Experienced with AWS, PostgreSQL, Node.js, and large-scale ETL. Seeking senior/staff engineering roles with AI-forward teams.

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Work Experience

SOFTWARE ENGINEER II **CONVEGENIUS** **JUN 2024 -PRESENT**

- Architected and owned end-to-end delivery of a fault-tolerant ETL pipeline ingesting **100M+** government education records daily across **6+ Indian states** — processing School, Student, and Employee datasets from **NIC UDISE+ APIs** using **parallel batch extraction** and PostgreSQL transactional staging with rollback, achieving **99.9%** data consistency across distributed sync operations.
- Designed and rearchitected a column-level **differential reconciliation engine** comparing **4–5 crore** rows × 80–90 columns across MIS and UDISE databases — identified **Materialized View** upsert limitations at **7 crore diff scale**, migrated to indexed PostgreSQL tables with a pre-aggregated summary layer, reducing processing from hours to minutes.
- Migrated **RBAC** and Field Master configuration layer from **DynamoDB** to **PostgreSQL** — redesigned schema, wrote migration verification scripts .
- Built **centralized log management** and observability layer across **6+ ETL pipelines** — replaced manual per-pipeline error file sharing with a **unified real-time dashboard** used by Engineering, Product, and QA, eliminating 30–60 min per-incident debug handoff across 3 teams.
- Led architecture-to-production delivery for government **MIS platform** serving 6+ Indian state education boards — owning technical design, **CI/CD** pipelines, and cross-functional alignment across Engineering, Product, QA, and state program managers to ensure contractual delivery milestones.
- Implemented production-grade recovery system using **exponential backoff** retry logic, **health-check endpoints**, AWS CloudWatch logging, and **Slack webhook** integration.

FULL-STACK DEVELOPER **TRAVEL CORPORATION OF INDIA** **MAY23-JUNE24**

- Spearheaded the **development** to **deployment** process of a customizable React theme-editor using **hooks**, **Redux**, and **Saga**, optimizing **asynchronous** operations to show itinerary details.
- Orchestrated the end-to-end implementation of Guide products across **backend**, **frontend**, and **mobile app** platforms, ensuring seamless **integration** and functionality.
- Integrated **AWS Cognito** for advanced **authentication** capabilities and developed **lambda functions** to incorporate bespoke authentication logic and user authentication.
- Engineered advanced **search algorithms** for guides, reducing **search times** by 2.35 seconds through **intelligent filtering** based on date, type, language, location, and booking status.
- Contributed to the enhancement of the guide mobile app built with **react-native**, refining user interface and experience design elements and publishing it on the **Play Store**.

SOFTWARE DEVELOPMENT ENGINEER **ARGOID [ACQUIRED BY AMAGI]** **JULY'22-MAY'23**

- Architected a reusable service layer for configuration, dashboard, and task & dependency management — reducing manual operational workload by 80%.
- Built an Administration Console with role-based access control (RBAC), enabling configuration management and real-time system monitoring.
- Developed a Shopify script for real-time behavioral data collection across e-commerce platforms using Google Tag Manager integrations.
- Optimized event server performance with JSON schema validation, increasing throughput to 50,000 requests/second.
- Built an image-to-text extraction pipeline using OCR, generating AI-powered product recommendations from extracted attributes.
- Engineered a dynamic HTML and CSS code generator using Angular, reducing UI customization time for internal teams.

SKILLS

- **Frontend:** React.js, Next.js, Angular, Redux, Redux-Saga, React Native, HTML5, CSS3
- **Backend:** Node.js, Express.js, Django, Dropwizard, Spring boot
- **Cloud & DevOps:** AWS Lambda, AWS DynamoDB, AWS S3, AWS CloudWatch, AWS Cognito, Cron Scheduling, CI/CD, Serverless Architecture, Snowflake, Redshift, EC2 , Elastic beanstalk

Projects

Chatter-A social-media server

1. **Functionalities:** Proper **authentication,** adding **posts**, **comments**, **likes**, **adding friends**, and **chatting engines** are implemented.
2. Proper **Database models** are added to handle data efficiently. Use **a message queue** to handle concurrent requests.
3. **Tech Stack Used:** MongoDB, NodeJS, Expressjs , JavaScript.
4. Link

Connector - A React app

1. Using the above **server API**, a **single-page web** app is built with **React**.
2. Proper **folder** structure and **code design implemented** which makes this web app **scaleable**.
3. Proper use of **custom hooks**, **providers**, **context API**, **state management** etc.
4. Github Link
5. Live Link

Coding Profile

1. **Leetcode Profile ->** Profile link

GitHub Profile

1. Link

EDUCATION

- **Bikaner Technical University Kota - B.Tech (Computer Science) 2018-2022**